

Programming Assignment 1

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ERP ID: 26106

Course: Introduction to Blockchain

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So, this is basically a document with all the OUTPUT PROOFS. My code file is attached too.

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PROOF 1: test.py got 40/40 marks

```
python testCode.py
Enter your erp id: 26106
Connected to backend
9889 9889 5
870957153513604955099783510612415718372330343087099090278033937281925992370 0
25710847192128860711870250207729231661178476158772968986907201458313275253420 0
40189016485892926282548271016069919390207554076965828391660538936131874139403 0
43558234995138531315276814253345165558232202230712793313817517100022236863201 0
61368317325581018485444567640877575581714997222663672800401752329642593681898 0
68957154921550612234866922340053330019704169724720129011905449946148775763014 0
85128033328377724622986738793818727043753474774787377171671560282379194970854 0
87316322514699652727249331442542001328835646615455141794091667247614728216777 0
97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000
99652275076723255094766834832004149220812205664595801229873801477016512014599 0
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 5000000000
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
```

```
61368317325581018485444567640877575581714997222663672800401752329642593681898 9999999835
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 0
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 7969690844
40189016485892926282548271016069919390207554076965828391660538936131874139403 8573843577
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 11524896630
68957154921550612234866922340053330019704169724720129011905449946148775763014 6426155594
85128033328377724622986738793818727043753474774787377171671560282379194970854 0
87316322514699652727249331442542001328835646615455141794091667247614728216777 0
97900424532147900438859704322823053922997230128479610197495596665802299846947 0
99652275076723255094766834832004149220812205664595801229873801477016512014599 5505410971
Bad transaction was trying to: 43558234995138531315276814253345165558232202230712793313817517100022236863201
870957153513604955099783510612415718372330343087099090278033937281925992370 1904043603
25710847192128860711870250207729231661178476158772968986907201458313275253420 2531522792
40189016485892926282548271016069919390207554076965828391660538936131874139403 9942931921
43558234995138531315276814253345165558232202230712793313817517100022236863201 1792825777
61368317325581018485444567640877575581714997222663672800401752329642593681898 3987997543
68957154921550612234866922340053330019704169724720129011905449946148775763014 60512845
85128033328377724622986738793818727043753474774787377171671560282379194970854 13396149551
87316322514699652727249331442542001328835646615455141794091667247614728216777 4671489268
97900424532147900438859704322823053922997230128479610197495596665802299846947 47783348
99652275076723255094766834832004149220812205664595801229873801477016512014599 11664420824
```

Your Marks: 40 out of 40

> python testCode.py

Enter your erp id: 26106

Connected to backend

9889 9889 5

870957153513604955099783510612415718372330343087099090278033937281925992370 0

25710847192128860711870250207729231661178476158772968986907201458313275253420 0

40189016485892926282548271016069919390207554076965828391660538936131874139403 0

43558234995138531315276814253345165558232202230712793313817517100022236863201 0

61368317325581018485444567640877575581714997222663672800401752329642593681898 0

68957154921550612234866922340053330019704169724720129011905449946148775763014 0

85128033328377724622986738793818727043753474774787377171671560282379194970854 0

87316322514699652727249331442542001328835646615455141794091667247614728216777 0

97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000

99652275076723255094766834832004149220812205664595801229873801477016512014599 0
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 5000000000
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 5000000000
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 5000000000
40189016485892926282548271016069919390207554076965828391660538936131874139403 5000000000
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 9999999835
68957154921550612234866922340053330019704169724720129011905449946148775763014 5000000000
85128033328377724622986738793818727043753474774787377171671560282379194970854 5000000000
87316322514699652727249331442542001328835646615455141794091667247614728216777 5000000000
97900424532147900438859704322823053922997230128479610197495596665802299846947 0
99652275076723255094766834832004149220812205664595801229873801477016512014599 5000000000
870957153513604955099783510612415718372330343087099090278033937281925992370 5000000000
25710847192128860711870250207729231661178476158772968986907201458313275253420 7969690844
40189016485892926282548271016069919390207554076965828391660538936131874139403 8573843577
43558234995138531315276814253345165558232202230712793313817517100022236863201 5000000000
61368317325581018485444567640877575581714997222663672800401752329642593681898 11524896630
68957154921550612234866922340053330019704169724720129011905449946148775763014 6426155594
85128033328377724622986738793818727043753474774787377171671560282379194970854 0
87316322514699652727249331442542001328835646615455141794091667247614728216777 0
97900424532147900438859704322823053922997230128479610197495596665802299846947 0

99652275076723255094766834832004149220812205664595801229873801477016512014599 5505410971

Bad transaction was trying to:

43558234995138531315276814253345165558232202230712793313817517100022236863201

870957153513604955099783510612415718372330343087099090278033937281925992370 1904043603

25710847192128860711870250207729231661178476158772968986907201458313275253420 2531522792

40189016485892926282548271016069919390207554076965828391660538936131874139403 9942931921

43558234995138531315276814253345165558232202230712793313817517100022236863201 1792825777

61368317325581018485444567640877575581714997222663672800401752329642593681898 3987997543

68957154921550612234866922340053330019704169724720129011905449946148775763014 60512845

85128033328377724622986738793818727043753474774787377171671560282379194970854 13396149551

87316322514699652727249331442542001328835646615455141794091667247614728216777 4671489268

97900424532147900438859704322823053922997230128479610197495596665802299846947 47783348

99652275076723255094766834832004149220812205664595801229873801477016512014599 11664420824

Your Marks: 40 out of 40

Then main.py was run, “rl” to fetch the latest chain of 696 blocks, and “validate” was run to validate the file, and “print” was done to print the chain. Then mining began.

PROOF 2: Mining was successful

> mine

Mining... this may take a while with difficulty 4

Still mining... tried up to nonce 50000

Still mining... tried up to nonce 100000

Still mining... tried up to nonce 150000

Still mining... tried up to nonce 200000

Still mining... tried up to nonce 250000

Still mining... tried up to nonce 300000

Still mining... tried up to nonce 350000

Still mining... tried up to nonce 400000

Still mining... tried up to nonce 450000
Still mining... tried up to nonce 500000
Still mining... tried up to nonce 550000
Still mining... tried up to nonce 600000
Still mining... tried up to nonce 650000
Still mining... tried up to nonce 700000
Still mining... tried up to nonce 750000
Still mining... tried up to nonce 800000
Still mining... tried up to nonce 850000
Still mining... tried up to nonce 900000
Still mining... tried up to nonce 950000
Still mining... tried up to nonce 1000000
Still mining... tried up to nonce 1050000
Still mining... tried up to nonce 1100000
Still mining... tried up to nonce 1150000
Still mining... tried up to nonce 1200000
Still mining... tried up to nonce 1250000
Still mining... tried up to nonce 1300000

block hash found

0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

Block successfully mined!

PROOF 3: chain successfully sent to backend

> send state

State sent to backend

PROOF 4: chain fetched and printed to see if my block was added

> rl

Requested longest chain, check pending_chains folder

Backend has no longer chain yet, please try again after some time...

> validate

No pending chains found to validate.

> print

... excluded al 686 blocks

Block index # 687

Transaction number 1276 with hash

106897296527924764199667125552980890037746170738505784713432458442547679971572

Transaction number 1279 with hash

4348696049650101035633769778528607889714584734871062526600404328859556934820

Transaction number 1283 with hash

15206552066005789095658745641384009236848085361028234329244152637804893412115

Transaction number 1290 with hash

98690495398309002924443911061725567558432810741273202135647905795356076794112

Transaction number 1295 with hash

46911183463424891503636513496414637039836375655325814063081513851381921458587

nonce: 1300460

previous_hash: 00004a3ec2f86ddc5b0bbe0573d23906e9adbef9d12a552d27b64472b145469c

hash: 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

Miner: 26106

PROOF 5: checked chain for my ERP using commands

> python -c "import glob,os,pickle; from FullNode import FullNode;

files=sorted(glob.glob('valid_chain/block*.block'), key=lambda x:int(os.path.basename(x)[5:-6])); f=[f for f in files

```
if os.path.basename(f)=='block687.block'[0]; b=pickle.load(open(f,'rb')); n=FullNode('26106'); print('index=',  
b.index); print('miner=', b.miner); print('hash=', n.computeBlockHash(b))"
```

index= 687

miner= 26106

hash= 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

```
> python -c "import glob,os,pickle; files=sorted(glob.glob('valid_chain/block*.block'), key=lambda  
x:int(os.path.basename(x)[5:-6])); found=[]; [found.append((b.index,b.miner)) for f in files for b in  
[pickle.load(open(f,'rb'))] if str(b.miner)=='26106']; print(found[-20:]); print('total matches =', len(found))"
```

[(687, '26106')]

total matches = 1

PROOF 6: Blocks added after my block

```
*****  
Block index # 687  
Transaction number 1276 with hash 106897296527924764199667125552980890037746170738505784713432458442547679971572  
Transaction number 1279 with hash 4348696049650101035633769778528607889714584734871062526600404328859556934820  
Transaction number 1283 with hash 15206552066005789095658745641384009236848085361028234329244152637804893412115  
Transaction number 1290 with hash 9869049539830900292443911061725567558432810741273202135647905795356076794112  
Transaction number 1295 with hash 46911183463424891503636513496414637039836375655325814063081513851381921458587  
-----  
nonce: 1300460  
previous_hash: 00004a3ec2f86ddc5b0bbe0573d23906e9adbef9d12a552d27b64472b145469c  
hash: 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3  
Miner: 26106  
*****  
  
*****  
Block index # 688  
Transaction number 1296 with hash 58987023161446331021402941306404632760026750423800979600783589351279961143360  
Transaction number 1298 with hash 8373493932771146940734093892759598869715673982991167397736691856870244968422  
Transaction number 1299 with hash 14291103965669749236736710049936839502406721666023804922010695810444918711410  
Transaction number 1300 with hash 78376476006371406786409599878621206075103688154040114009748082934149395746142  
Transaction number 1302 with hash 98625460474457581848192473545597839122007619385678105446656565126418251611628  
-----  
nonce: 497270  
previous_hash: 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3  
hash: 0000aade5e23a1f5e5be8a1a531fa5c7c3ba06dfad395e2c97f43d20c2e40bf  
Miner: 26956  
*****  
  
*****
```

Block index # 686

Transaction number 1263 with hash

85411459801750583924379771724100603743214712645587652217657376285791888971269

Transaction number 1264 with hash

113608330279527712129199580997234049119827061751021510932869204903405850888594

Transaction number 1266 with hash
108026490070220948258683347093258526452013000821197448411100869416388837729223

Transaction number 1272 with hash
84829507718691180729857441125589395114242926417469530439318821189967110921051

Transaction number 1275 with hash
100077431043935180347839712544331806663154787643642819051867774084369449362715

nonce: 104640

previous_hash: 00009b1f8dde3823f13d470df1d26b4b76eb4c14a98b451ef98698147c25f719

hash: 00004a3ec2f86ddc5b0bbe0573d23906e9adbef9d12a552d27b64472b145469c

Miner: 28991

Block index # 687

Transaction number 1276 with hash
106897296527924764199667125552980890037746170738505784713432458442547679971572

Transaction number 1279 with hash
4348696049650101035633769778528607889714584734871062526600404328859556934820

Transaction number 1283 with hash
15206552066005789095658745641384009236848085361028234329244152637804893412115

Transaction number 1290 with hash
98690495398309002924443911061725567558432810741273202135647905795356076794112

Transaction number 1295 with hash
46911183463424891503636513496414637039836375655325814063081513851381921458587

nonce: 1300460

previous_hash: 00004a3ec2f86ddc5b0bbe0573d23906e9adbef9d12a552d27b64472b145469c

hash: 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

Miner: 26106

Block index # 688

Transaction number 1296 with hash

58987023161446331021402941306404632760026750423800979600783589351279961143360

Transaction number 1298 with hash

8373493932771146940734093892759598869715673982991167397736691856870244968422

Transaction number 1299 with hash

14291103965669749236736710049936839502406721666023804922010695810444918711410

Transaction number 1300 with hash

78376476006371406786409599878621206075103688154040114009748082934149395746142

Transaction number 1302 with hash

98625460474457581848192473545597839122007619385678105446656565126418251611628

nonce: 497270

previous_hash: 0000c88055d80223236e80abbffc460a381b0a5e0d492ca30f722e9ccf74b1b3

hash: 0000aade5e23a1f5e5be8a1a531fa5c7c3cba06dfad395e2c97f43d20c2e40bf

Miner: 26956

Block index # 689

Transaction number 1308 with hash

31759150156224767616442148047435086625994287837022129046828002942296729197176

Transaction number 1309 with hash

56632493402093467299129210384003882649703508759723019171271524851781926133904

Transaction number 1314 with hash

14291353302511323610882808553071384971982812012778503574391875536993893082315

Transaction number 1316 with hash

55410992612947734630846407697823499849137451586003889981216754956938491180134

Transaction number 1319 with hash

51205624483406576833616732064705841873609002274412221812223173010432674767866

nonce: 249030

previous_hash: 0000aade5e23a1f5e5be8a1a531fa5c7c3cba06dfad395e2c97f43d20c2e40bf

hash: 00007425b5be884130e621a70af6ce4baf0ff11c72adb026fbb4ae688ee69ead

Miner: 26906

>

so after doing “rl” and “validate” after a couple of hours, we see that more blocks have been added after my block from different ERPs meaning my block has been accepted by the system and is included in the longest chain successfully.